

Five-Page Proof Skeleton

External-audit version

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1. Mass-action and conservation

At a positive equilibrium, define $v_r = k_r(x^*)^{y_r}$ and $h_i = (x_i^*)^{-1}$. Direct differentiation gives

$$J = \Gamma \operatorname{diag}(v) Y^T \operatorname{diag}(h), \quad v > 0, \quad \Gamma v = 0,$$

and every such pair is realized by $x_i^* = h_i^{-1}$ and $k_r = v_r/(x^*)^{y_r}$. No deficiency or weak-reversibility assumption is used.

Put $S = \operatorname{im} \Gamma$. Since $J(\mathbb{R}^n) \subseteq S$, if $J|_S$ is Hurwitz then $J(S) = S$, $S \cap \ker J = 0$, and rank-nullity gives $\mathbb{R}^n = S \oplus \ker J$. Thus conservation zeros are semisimple and

$$\det(\lambda I - J) = \lambda^{n - \dim S} \det(\lambda I_S - J|_S) > 0 \quad (\lambda > 0).$$

A nonconstant Neumann eigenfunction has zero mean. Hence every species amplitude preserves integrated conservation laws; unequal diffusion generally does not preserve S , so the nonzero-mode matrix is tested on $\mathbb{R}^n$.

Choose a rational basis B of S and left inverse C . The reduced matrix $R = CJB$ represents $J|_S$. Replacing B by BT conjugates R by T , so Routh–Hurwitz inequalities are coordinate-independent.

2. Fixed-J theorem and the network iff

For $D = \text{diag}(d)$, multilinearity gives

$$\det(D - J) = \sum_{I \subseteq [n]} (-1)^{|I|} \det J[I, I] \prod_{j \notin I} d_j.$$

If every signed principal minor is nonnegative, replacing d_j by $d_j + \lambda$ makes this determinant strictly positive for every $\lambda > 0$; no positive real eigenvalue of $J - D$ is possible.

If one coefficient indexed by I is negative, set $d_j = t$ off I and $d_j = t^{-1}$ on I . Its exponent exceeds the exponent of the K -term by $|I \triangle K|$, so it uniquely dominates for large finite t . Then $\det(D - J) < 0$; the monic characteristic polynomial of $J - D$ is negative at zero and positive at large positive argument, yielding a positive real root. For Hurwitz J , the empty and full coefficients are positive.

Now write $A(v) = \Gamma \text{diag}(v) Y^T$. Principal column scaling gives

$$(-1)^{|I|} \det(A(v) \text{diag}(h))[I, I] = \left(\prod_{i \in I} h_i \right) (-1)^{|I|} \det A(v)[I, I].$$

Combining the parametrization, relative Hurwitz test, and fixed-J theorem gives the exact network iff.

For existential-real membership, Boolean selectors satisfy

$$\det(I - S_b - S_b A(v) S_b) = (-1)^{|I|} \det A(v)[I, I].$$

Rational bases have polynomial encoding length, and determinant, characteristic-coefficient, and Hurwitz circuits have polynomial size.

3. Open-cube family and row splitting

For a ‘PARTITION’ input, pad to $a \in \mathbb{Z}^m$ with $m = k^2$, and set

$$Q = I + aa^T, \quad \beta = 1 - \frac{1}{2m(1 + a^T a)}, \quad B(x, y) = \begin{pmatrix} -kQ & y \\ x^T & k\beta \end{pmatrix}.$$

A partition sign vector t gives $x = rt$, $y = -rt$ with $r = (1 + \beta)/2$. Orthogonal decomposition leaves one 2×2 block whose trace is negative and determinant $k^2(r^2 - \beta) > 0$.

Conversely, interpolate from $(0, 0)$, which has exactly one positive eigenvalue, to a Hurwitz endpoint. An interior singular point satisfies

$$\theta^2 x^T Q^{-1}(-y) = m\beta.$$

Since $\theta < 1$, the cube maximum exceeds $m\beta$. Positive definiteness and Sherman–Morrison give

$$\max_{x, z \in [-1, 1]^m} x^T Q^{-1} z = m - \frac{\min_{t \in \{\pm 1\}^m} (a^T t)^2}{1 + a^T a}.$$

The strict half-unit gap forces the integer $a^T t$ to be zero.

Write $B(q) = B_0 + UX(q)$ and define

$$L(q) = \begin{pmatrix} B_0 & U \\ X(q)(B_0 + I) & X(q)U - I \end{pmatrix}.$$

Direct multiplication gives

$$\begin{pmatrix} I & 0 \\ -X & I \end{pmatrix} L(q) \begin{pmatrix} I & 0 \\ X & I \end{pmatrix} = \begin{pmatrix} B(q) & U \\ 0 & -I \end{pmatrix}.$$

Every varying row of L contains exactly one scalar parameter.

4. Scaling elimination and mass-action realization

Suppose $L(q)D$ is Hurwitz for some arbitrary positive diagonal D . At $q = 0$, block triangularity shows that $L(0)D$ has exactly one positive eigenvalue: the $-kQD_t$ block is similar to a negative definite matrix, one scalar block is positive, and the added block is negative. The signed determinant therefore changes sign along $L(\theta q)D$. At an interior singularity, nonsingularity of D and $\det L(q) = \det B(q)$ transfer singularity to $B(\theta q)$. The strict open-cube argument then forces a partition. This eliminates arbitrary relative right scaling only for the constructed family.

For a variable row $a + qb$, clear denominators by c and choose an integer base complex y^0 so that $y^\pm = y^0 + c(a \pm b)$ are nonnegative. Use reactions

$$y^+ \rightarrow y^+ + e_i, \quad y^- \rightarrow y^- + e_i, \quad y^0 \rightarrow y^0 - e_i.$$

Every positive steady flux is $(u, w, u + w)$ and contributes

$$c((u + w)a + (u - w)b) = \rho(a + qb), \quad q = \frac{u - w}{u + w} \in (-1, 1).$$

Every $\rho > 0, q \in (-1, 1)$ is attained. Fixed rows use a two-reaction pair. All flux constraints separate by row, proving equality of the complete image.

Thus every homogeneous Jacobian is $RL(q)H$, similar to $L(q)HR$. A fixed positive diagonal entry creates a negative signed singleton minor whenever the Jacobian is Hurwitz. The network is admissible exactly for YES partition instances. Counts are $3m + 1$ species and $8m + 2$ reactions. Molecularities are unbounded.

5. Fixed n , certificates, and audit boundary

Let $K = \{v \geq 0 : \Gamma v = 0\}$ and $P = \Phi(K)$ with $\Phi(v) = \Gamma \text{diag}(v)Y^T$. A strict positive flux implies $\text{ri } K = K \cap \mathbb{R}_{>0}^m$; linear maps preserve relative interiors, so positive factors are exactly $\text{ri } P$.

Every extreme ray of K is a support-minimal positive dependency and has support at most $\text{rank } \Gamma + 1 \leq n + 1$. For fixed n , enumerate all bounded supports, compute exact minor generators, map them to P , and in fixed dimension compute span, lineality, and relative facets. The remaining formula has $n^2 + n$ real variables (plus one for designated mobility), fixed degree, and polynomial coefficient data. Fixed-variable real-algebraic decision is polynomial in bit complexity.

For certificates, equationize strict signs by $gz^2 - 1 = 0$ or $(-g)z^2 - 1 = 0$. A YES branch has a real-algebraic sample point. An infeasible branch has a Real Nullstellensatz identity

$$-1 = \sum_j s_j^2 + \sum_k q_k F_k.$$

These objects are finite and exactly checkable; no polynomial size or practical general generator is claimed.

The audit attacked the three critical steps independently: arbitrary right scaling, equality of the entire reaction row image, and fixed-dimensional bit complexity. It found no central defect. The result remains limited to weak stationary linear instability, weak NP-hardness, and unbounded-molecularity hardness; external priority and peer review remain necessary.